Egg segmentation and tracking

Computer Vision Intern Task | reproducible experimental report

Executive summary

A pretrained MobileSAM model produced reference egg masks for all 110 images. A classical local graph-cut method, selected on the development set, achieved mean IoU **0.532** and Dice **0.627** on 22 evaluation images. A two-ID tracker processed the complete 60-second reference video and marks uncertain reidentification after occlusion.

Dataset and split

Roboflow egg v1 classification export: 110 images (108 in the egg folder and 2 Unlabeled), 640 x 640, CC BY 4.0. The supplied ZIP is v1; the online v2/v3 pages list 160 images but were not the provided local export. The exact 4:1 split has **88 development and 22 evaluation** images. Source IDs, pHash, HSV histograms and geometrically verified ORB matches form 99 scene groups; no group crosses the split.

Leakage audit: 0 exact duplicate files across partitions; minimum cross-split pHash distance 16; 0 cross-split pairs satisfy the grouping rules. A manual check caught related captures 151/152 and pot scenes 113/135; both pairs now share a partition. These rules reduce leakage but cannot prove that every capture session is independent.



The provided images span hands, ash, textured flooring and patterned fabric.

Task 1 - pretrained reference segmentation

MobileSAM automatic proposals at 384 px were filtered by area, shape, border contact and overlap. Visual audit found missed eggs, so 23 images received human-selected positive prompts; four difficult images used boxes. Prompts were chosen from the image, independent of classical predictions. Two boiling-water images have no visible egg mask. All masks, prompt coordinates and model checksum are saved. These are **pseudo-labels**, not human ground truth; agreement with them does not prove absolute accuracy.

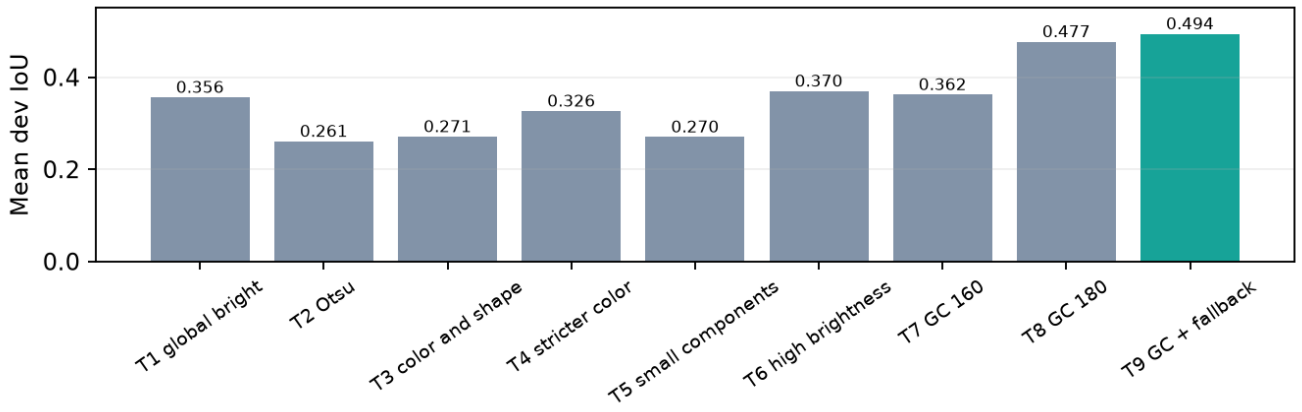
Task 2 - classical method

The final method converts each image to grayscale and HSV, selects bright low-saturation seed regions, removes tiny noise, filters implausible components by shape, and runs local GrabCut around each seed. GrabCut expands seeds toward full egg boundaries. If it finds no candidate at all, a conservative global brightness mask is used. That fallback was chosen on development images only. The video uses a separate documented brightness setting and table ROI; neither classical pipeline uses MobileSAM at runtime.

Experiment log

Each trial used the same 88-image development set. Parameters and reasons are in outputs/experiment_log.json; the table below gives the decision path. The selection rule was highest mean dev IoU.

Trial	Change / reason	Dev IoU	Decision
T1 global bright	Simple light-egg baseline	0.356	Drop
T2 Otsu	Adaptive global threshold for varying exposure	0.261	Drop
T3 color and shape	Reject colored background and non-elliptical regions	0.271	Drop
T4 stricter color	Reduce bright clutter false positives	0.326	Drop
T5 small components	Recover distant eggs	0.270	Drop
T6 high brightness	Test stronger foreground contrast	0.370	Drop
T7 graph cut 160	Grow bright egg cores to full boundaries using local color models	0.362	Drop
T8 graph cut 180	Stricter seeds before local graph cut	0.477	Drop
T9 graph cut with empty-mask fallback	Use a conservative global mask only when graph cut finds no egg	0.494	Keep

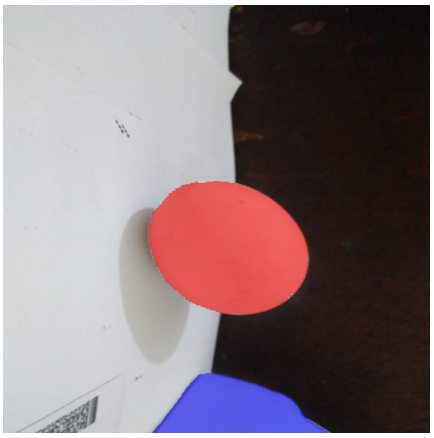


Task 3 - grouped evaluation

On the fixed 22-image grouped split, the prior graph-cut method scored IoU **0.509**; the dev-selected fallback scored **0.532** (image-bootstrap 95% interval 0.394-0.663). Dice 0.627; pixel precision 0.684; pixel recall 0.682. Pixel accuracy is 0.910 and balanced accuracy is 0.813; predicting background everywhere would already achieve 0.872 pixel accuracy. Object matching at IoU ≥ 0.5 yields precision 0.598 and recall 0.554; mean absolute count error 0.95. Metrics are per-image means against MobileSAM pseudo-labels.

Qualitative error analysis

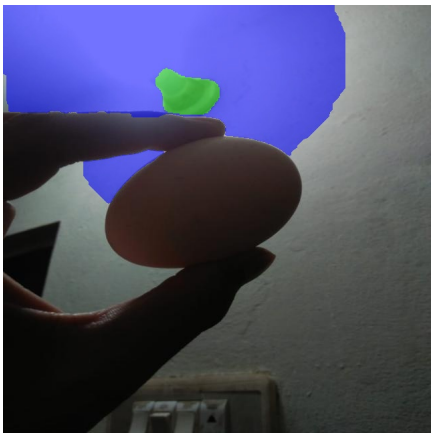
Green = overlap; red = MobileSAM only; blue = classical method only. Samples are the lowest-IoU evaluation cases, chosen by a fixed ranking. Open outputs/explorer/index.html for the full interactive gallery.



not_damaged_30_jpg
IoU 0.000



not_damaged_36_jpg
IoU 0.000



not_damaged_24_jpg
IoU 0.051

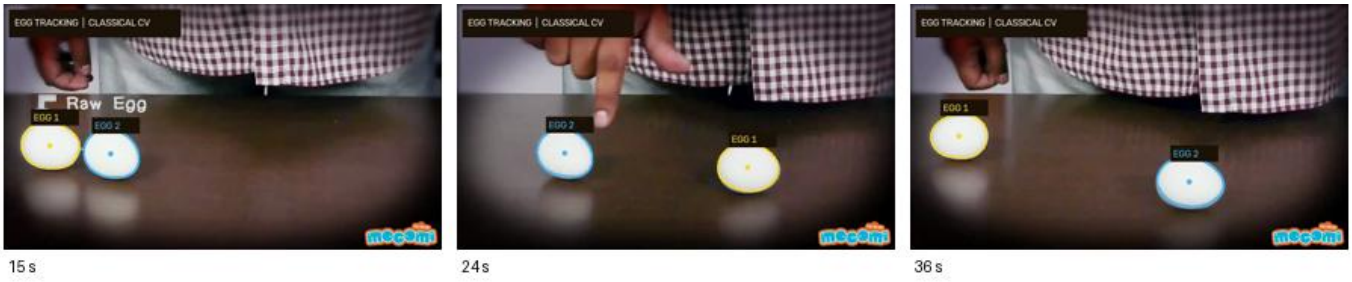
The lowest-scoring cases expose missed eggs, bright-background false positives and boundary spill. The ranking is generated from per-image scores, so it remains correct if the split or selected method changes.

Observed gaps

Dark or shadowed eggs can fail the bright-seed step. Bright cookware, tissue and reflections can become false positives. GrabCut can absorb a similar-colored background or split an egg occluded by a rail. The reference pipeline itself misses partially submerged eggs and sometimes selects non-egg proposals; the saved QC sheet and manual prompt log make this visible. Next improvements: human instance masks for a representative subset, illumination normalization, temporal consistency for video, and a dedicated egg detector to provide proposals.

Video tracking and reproducibility

Task 4 uses a table ROI and a stricter brightness seed to suppress hands, captions and reflections. Connected components are assigned to two persistent tracks with global minimum-distance matching; a frame gap over 24 frames or a jump over 100 px is marked uncertain. Across 1,440 video frames, the output contains 1351 egg detections and 3 uncertain reidentification events. Identity during hand occlusion cannot be established confidently from two visually similar eggs; these events are explicitly signalled rather than silently relabelled.



Three frames from the annotated output. IDs remain distinct; the video marks uncertain recovery with ?.

Tracking video

Public Google Drive video: https://drive.google.com/file/d/1oLbEB3EDDzFkoCDZdwqK_lTCbqnHanMv/view?usp=drivesdk

Reproduce

Create a Python 3.12 environment, install requirements.txt, keep the supplied egg.v1i.folder at the repository root, and run scripts/generate_reference.py, scripts/repair_reference.py, scripts/run_experiments.py, scripts/track_video.py, and scripts/make_report.py in that order. See README.md for exact commands. The MobileSAM checkpoint is downloaded separately and identified by checksum in outputs/reference_provenance.json.

Attribution and AI use

Dataset: Fire, egg v1, Roboflow Universe, CC BY 4.0, https://universe.roboflow.com/fire-bf7qe/egg-vzhtr/dataset/1 . Model: Ultralytics MobileSAM, https://docs.ultralytics.com/models/sam/ . Source video: Mocomi, https://www.youtube.com/watch?v=Y4JyV94y2_A . AI coding assistance was used for code drafting, debugging, and report wording; the workflow, visual QC, method comparison, and interpretation are documented for independent review.

Limitations

The dataset has classification folders, not human segmentation annotations. Earlier split results were seen before scene grouping was strengthened, so this revised evaluation is exploratory rather than a pristine untouched test. A future independent capture-session split with human instance masks would give stronger accuracy evidence. Scene grouping cannot eliminate every visual relationship, and the video-specific ROI is tailored to this clip.